Annotation Transfer Tool

https://github.com/aiyshwary/Annotation-Transfer-Tool

Python · Computer Vision · AWS S3 · CVAT · PyTorch

OVERVIEW

A smart auto-annotation pipeline that leverages deep image similarity to transfer labels from previously human-annotated images to new unannotated ones, significantly reducing manual labeling effort in computer vision workflows.

IMPACT

Reduced manual annotation effort by automatically transferring labels from existing human-labeled images to new images using deep feature similarity and a modular 8-step pipeline integrated with CVAT and AWS S3.

TECHNICAL WALKTHROUGH

This project is an Annotation Transfer Tool that significantly reduces manual labeling effort by automatically transferring annotations from previously human-labeled images to new images using deep image similarity.

In short

i) Find similar images → reuse their annotations → verify visually → export back to CVAT

What problem does it solve?

Manual annotation is slow and expensive. Even auto-annotation models struggle when data is limited or domain-specific. This tool leverages already annotated images and transfers their labels to new images by identifying visually similar samples using deep learning.

Core idea

Instead of predicting bounding boxes from scratch, the tool searches for visually similar images using deep feature embeddings and transfers annotations from the closest matches.

This makes it

i) Faster than manual labeling

ii) More reliable than blind auto-annotation

iii) Especially effective in low-variation industrial datasets

End-to-end pipeline explanation

Configuration-driven setup

The entire pipeline is controlled through a single config.yml, which defines input paths, S3 locations, and processing parameters.

This makes the pipeline

- **i)** Reproducible
- **ii)** Easy to reconfigure across datasets

Image & annotation ingestion
Images are downloaded from S3 based on a Datumaro JSON file exported from CVAT. Only relevant images are pulled, avoiding unnecessary data transfer.

Scripts involved

- **i)** s3_download.py
- **ii)** datuamaru_crop_sort.py

Label normalization
Annotations are normalized into a consistent format so that downstream processes like cropping and similarity matching work reliably.

Script

- **i)** normalize_labels.py

Crop, mask, and label generation
For each annotated object, the tool generates image crops, corresponding label files, and binary masks. This creates a reference library of labeled object appearances.

- **i)** crops_generation.py

Deep image similarity search
A deep learning–based feature extractor generates embeddings for image crops. Using DeepImageSearch, the tool identifies visually similar images across the dataset.

- **i)** CheckSimilarityScore_gpu.py

This is where

- **i)** Feature-based search replaces brute-force labeling
- **ii)** Previously labeled examples guide new annotations

Annotation transfer & correction
Annotations from the most similar matches are transferred to the target images. Any mismatches or incorrect labels are corrected and mapped back to the original annotation files.

i) Correcting_matching.py

This step ensures

i) Label consistency

ii) Reduced noise from similarity errors

Visualization for validation
Both normalized and denormalized annotations are visualized to verify correctness before final export.

Scripts

i) visualization_normalized.py

ii) visualization_denormalized.py

Export back to CVAT
Finally, the updated annotations are converted back into Datumaro JSON format, making them directly uploadable to CVAT.

i) fetch_json_data.py

Two ways to run the system

1) Full automation

The entire pipeline can be run with a single command (run.py) for batch processing.

2) Step-by-step execution

Each stage can also be executed independently for debugging, validation, or partial runs.

Interactive demo
A Jupyter notebook (DEMO.ipynb) demonstrates the entire workflow interactively, making it easy to understand intermediate outputs and debug results.
I built a deep similarity–based annotation transfer system that automatically reuses existing human annotations to label new images, reducing manual labeling effort while maintaining annotation quality.

KEY DETAILS

1. Downloads labeled images from AWS S3 and extracts annotations from Datamaro JSON format.

2. Generates image crops, normalizes labels, and creates masks for each labeled region.

3. Uses DeepImageSearch (GPU-accelerated) to identify the most visually similar labeled images for each new image.

4. Transfers and corrects annotations from matched images, then exports back to CVAT-compatible JSON.

5. Includes normalized and denormalized visualization scripts and an interactive Jupyter demo notebook.

6. Uses VGG19 as the feature extraction backbone with FAISS indexing for similarity search; supports incremental indexing so previously indexed images are not re-processed.

7. Annotation transfer selects the best match per crop by minimum cosine distance, replacing class and attribute tokens while preserving original polygon coordinates.

8. Automates CVAT task creation via REST API — programmatically creates cloud storage connections, generates S3-compatible JSONL manifests with per-image checksums, and uploads Datumaro ZIP annotations.